

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. INTRODUCTION:

To address the limitations of manual graphic design and the narrative shortcomings of standalone AI image generators, we propose **ComicCraft**. This comprehensive web platform acts as a fully automated digital comic publisher, integrating Large Language Models (LLMs) with advanced text-to-image AI to streamline creative storytelling.

3. SYSTEM OVERVIEW:

Built upon the high-performance FastAPI framework, the proposed system offers a frictionless user experience. Through a streamlined HTML interface, users can define their core narrative (e.g., Prompt, Lead Character, Tone, Art Style). Upon form submission, the backend orchestrates a multi-step pipeline: outlining the story, writing the script, rendering the images, displaying a web preview, and generating an offline document.

4. CORE CAPABILITIES

- **Abstracted User Interface:** A Jinja2-powered frontend that hides the complex prompt-engineering processes from the end-user.
- **Dual-LLM Architecture:** Employs two variants of Google Gemini—one optimized for structural JSON logic and the other for rich, conversational dialogue generation.
- **Dynamic Visual Rendering:** Seamlessly translates AI-written scene descriptions into API calls for Hugging Face Stable Diffusion, producing vibrant comic panels.
- **Interactive Web Layout:** A dynamically generated preview page that loops through the backend data to present the comic intuitively on the browser.

- **Automated Document Export:** A one-click PDF generation feature that binds the completed story and imagery into a shareable file using the FPDF library.

5. END-USER WORKFLOW

1. The user navigates to the ComicCraft homepage.
2. The user fills in their story parameters and submits the form.
3. The system processes the AI requests in the background.
4. The generated comic is displayed panel-by-panel on the screen.
5. The user downloads the final creation as a formatted PDF.

6. KEY BENEFITS:

The proposed solution dramatically accelerates the creative process, reducing the comic production timeline from several weeks to less than a minute. It offers immense scalability, supports various artistic genres, and makes visual literature accessible to everyone regardless of their drawing abilities.